CSE 717 — Information Security

Topic 5: Game Theory & Nash Equilibrium — Payoff Matrix, Dominant Strategy

Complete Past Paper Solutions: 2020–2024

Source: Lc#3 (Game Theory in Network Security)

2/5 years (2023, 2024 – both most recent), 3–6 marks. Cheap, mechanical marks once the method is drilled.

Contents

Quick Reference — Finding Nash Equilibrium (CHEAT SHEET)

2020–2022: no Game Theory question found

This topic is NEW to the 2026 syllabus pattern — only 2023 and 2024 (the two most recent years) ask it, both as small (3-mark) parts of a larger Section-B question. Don't over-invest, but the method below is fast and mechanical once memorized.

Key definitions (Lc#3)

- **Player:** an entity participating in the game (e.g., Alice, Bob, Eve, Mallory).
- **Action / Strategy:** a choice a player can make.
- **Payoff:** the gain (or loss) a player receives, depending on *both* players' choices.
- **Payoff matrix:** a table where each cell (x, y) shows (Row player's payoff, Column player's payoff) for that combination of actions.
- **Nash Equilibrium (NE):** a combination of strategies where **neither player can improve their own payoff by unilaterally changing their own strategy**, given the other player's choice. A game can have 0, 1, or multiple NEs.
- **Strictly dominant strategy:** a strategy that gives a player a **strictly higher payoff no matter what the other player does** — i.e., it is the best choice regardless of the opponent's action.

The 6-Step "Underline" Method to Find All NEs (Lc#3)

1. Pretend you are Player 1 (row player).
2. Assume Player 2 (column player) picks a particular action (fix a column).

3. Looking down that column, find Player 1's **best payoff** (highest number) and underline it.
4. Repeat Steps 2–3 for Player 2's other action(s) (the other column(s)).
5. Now switch roles: for each row (Player 1's action fixed), find Player 2's best payoff across that row and underline it.
6. **Any cell where BOTH numbers are underlined is a Nash Equilibrium.**

Dominant strategy check: for each player, compare their payoffs for a fixed strategy across all of the opponent's actions. If one strategy gives a strictly higher payoff in *every* case, it is strictly dominant.

Worked example from slides — 2×2 payoff matrix

	Action C	Action D
Action A	$10, 2$	$8, 3$
Action B	$12, 4$	$10, 1$

Underline pass: If P2 plays C, P1's best is B (12>10) → underline 12. If P2 plays D, P1's best is B (10>8) → underline 10. If P1 plays A, P2's best is D (3>2) → underline 3. If P1 plays B, P2's best is C (4>1) → underline 4.
⇒ Cell $(B, C) = (12, 4)$ has BOTH numbers underlined ⇒ **unique NE** $= (B, C)$.
Also, P1's best response is always B (12>10 and 10>8 in both columns) ⇒ **P1 has a strictly dominant strategy: B**.
Second slide example (coordination game) can have **two** NEs simultaneously, e.g. $(\text{Yes}, \text{Yes}) = (20, 20)$ and $(\text{No}, \text{No}) = (10, 10)$ — both are NEs even though $(20, 20)$ is more efficient; game theory alone cannot predict which occurs.

2024 — Section B Q7(a): Eve/Mallory PT vs ND payoff matrix — find all NE, dominant strategies (3)

Question — 2024 Section B Q7(a)

Eve and Mallory are choosing which cybersecurity training to attend: Penetration Testing (PT) or Network Defense (ND). Their satisfaction (payoff) depends on both their choices; they prefer attending the same training together, but each has slightly different preferences. The payoff table shows (Eve's payoff, Mallory's payoff):

	Mallory: PT	Mallory: ND
Eve: PT	$4, 3$	$1, 2$
Eve: ND	$2, 1$	$3, 4$

(i) Identify all Nash equilibria in this scenario.

(ii) Does Eve have a strictly dominant strategy? Does Mallory? [3]

Answer — 2024 Section B Q7(a)

Step 1–2 — if Mallory plays PT, compare Eve's payoffs down the PT column: 4 (Eve:PT) vs 2 (Eve:ND). Best $= 4 > 2 \rightarrow$ underline Eve's 4 in cell (PT,PT).
If Mallory plays ND, compare Eve's payoffs down the ND column: 1 (Eve:PT) vs 3 (Eve:ND). Best $= 3 > 1 \rightarrow$ underline Eve's 3 in cell (ND,ND).
If Eve plays PT, compare Mallory's payoffs across the PT row: 3 (Mallory:PT) vs 2 (Mallory:ND). Best $= 3 > 2 \rightarrow$ underline Mallory's 3 in cell (PT,PT).
If Eve plays ND, compare Mallory's payoffs across the ND row: 1 (Mallory:PT) vs 4 (Mallory:ND). Best $= 4 > 1 \rightarrow$ underline Mallory's 4 in cell (ND,ND).

	Mallory: PT	Mallory: ND
Eve: PT	$\underline{4}, \underline{3}$	$1, 2$
Eve: ND	$2, 1$	$\underline{3}, \underline{4}$

(i) Nash Equilibria: both cells $(\text{PT}, \text{PT}) = (4, 3)$ and $(\text{ND}, \text{ND}) = (3, 4)$ have BOTH numbers underlined.

$$\boxed{\text{NE}_1 = (\text{Eve:PT}, \text{Mallory:PT}) = (4, 3), \qquad \text{NE}_2 = (\text{Eve:ND}, \text{Mallory:ND}) = (3, 4)}$$

This is a **coordination game** — both players prefer to attend the same training, but disagree on which one (Eve prefers PT, Mallory prefers ND).
(ii) Dominant strategy check:

- **Eve:** if Mallory plays PT, Eve's best is PT (4>2); if Mallory plays ND, Eve's best is ND (3>1). Eve's best choice **depends on Mallory's choice ⇒ Eve has NO strictly dominant strategy.**
- **Mallory:** if Eve plays PT, Mallory's best is PT (3>2); if Eve plays ND, Mallory's best is ND (4>1). Mallory's best choice also **depends on Eve's choice ⇒ Mallory has NO strictly dominant strategy either.**

2023 — Section B Q8(a): Alice/Bob ST#1 vs ST#2 payoff matrix — NE + dominant strategy (3)

Question — 2023 Section B Q8(a)

Alice and Bob develop two new network systems, but they forgot to agree on which security strategy to follow: Strategy 1 (ST#1) or Strategy 2 (ST#2). Their payoffs depend on which strategy they choose and whether they both apply the same one. Alice chooses a row, Bob chooses a column, and Alice's payoff is listed first:

	Bob: ST#1	Bob: ST#2
Alice: ST#1	$10, 10$	$8, 4$
Alice: ST#2	$4, 8$	$5, 5$

(i) Find the Nash equilibrium of this scenario, if any.

(ii) Does Alice have a strictly dominant strategy? Does Bob? [3]

Answer — 2023 Section B Q8(a)

If Bob plays ST#1, Alice's payoffs: 10 (ST#1) vs 4 (ST#2). Best $= 10 > 4 \rightarrow$ underline Alice's 10.
If Bob plays ST#2, Alice's payoffs: 8 (ST#1) vs 5 (ST#2). Best $= 8 > 5 \rightarrow$ underline Alice's 8.
If Alice plays ST#1, Bob's payoffs: 10 (ST#1) vs 4 (ST#2). Best $= 10 > 4 \rightarrow$ underline Bob's 10.
If Alice plays ST#2, Bob's payoffs: 8 (ST#1) vs 5 (ST#2). Best $= 8 > 5 \rightarrow$ underline Bob's 8.

	Bob: ST#1	Bob: ST#2
Alice: ST#1	$\underline{10}, \underline{10}$	$\underline{8}, 4$
Alice: ST#2	$4, \underline{8}$	$5, 5$

(i) Nash Equilibrium: only cell $(\text{ST\#1}, \text{ST\#1}) = (10, 10)$ has BOTH numbers underlined.

$$\boxed{\text{NE} = (\text{Alice:ST\#1}, \text{Bob:ST\#1}) = (10, 10)} \quad \text{— unique NE}$$

(ii) Dominant strategy check:

- **Alice:** if Bob plays ST#1, ST#1 (10) > ST#2 (4). If Bob plays ST#2, ST#1 (8) > ST#2 (5). **ST#1 is strictly better in BOTH cases ⇒ Alice has a strictly dominant strategy: ST#1.**
- **Bob:** if Alice plays ST#1, ST#1 (10) > ST#2 (4). If Alice plays ST#2, ST#1 (8) > ST#2 (5). **ST#1 is strictly better in BOTH cases ⇒ Bob also has a strictly dominant strategy: ST#1.**

This is the cleanest possible outcome: both players have a dominant strategy, both dominant strategies coincide, and the resulting cell is the unique NE — a "dominant strategy equilibrium".

2023 — Q4(b): Design a game-theory strategy to identify attackers in a mobile social network (3)

Question — 2023 Q4(b)

Using game theory, design a strategy to identify attackers in a mobile social network. [3]

Answer — 2023 Q4(b) (Lc#3 "Identifying Attackers in a Mobile Social Network")

Model the network as a game between two players:

- **Player 1 — the Server** (network/security monitor), which connects with each node.
- **Player 2 — a Node**, which can be either **benign** (a normal user) or **malicious** (an attacker) — the server does not directly know which.

Actions available to each player:

- **Server's actions:** (1) Do nothing, (2) Send the node a normal packet, (3) Set up surveillance on the node.
- **Node's actions:** (1) Do nothing / ignore, (2) Forward the packet normally, (3) Damage / drop the packet (malicious behavior).

Payoff structure (the strategy design):

- If the server **sends a normal packet**: good outcome if the node is **benign** (forwards it normally), bad outcome if the node is **malicious** (damages it).
- If the server **surveils**: good outcome if the node is **malicious** (caught/detected), but imposes a cost/bad outcome if the node was actually **benign** (wasted resources, degraded service).
- If the server **does nothing**: safe but no information gained — a malicious node could infiltrate undetected.

The strategy: the server should choose a **mixed strategy** — i.e., randomize between "send normal traffic" and "surveil" with some probability for each node, rather than a fixed rule. This is because:

1. Always surveilling everyone ⇒ degraded service for all benign (majority) nodes.
2. Never surveilling ⇒ malicious nodes infiltrate freely.
3. By analyzing the **payoff matrix** of (server action × node type/action) and finding the **Nash equilibrium** of this game, the server can compute the **optimal surveillance probability** that balances:
    - maximizing detection of malicious nodes (security), against
    - minimizing disruption to benign nodes (quality of service).

⇒ **Conclusion:** game theory provides a mathematical framework to automate this trade-off analysis at scale (*"analyze hundreds of thousands of 'what ifs'"*), giving network administrators a principled, equilibrium-based surveillance policy instead of an ad-hoc one.

2020–2022 — No Game Theory question found

2020/2021/2022 papers

No Game Theory / Nash Equilibrium / payoff-matrix question was identified in the 2020, 2021, or 2022 past papers. This is a topic that appears to have been newly introduced into the syllabus for 2023+2024 (both most recent years) — a strong recency signal that it will likely repeat in 2026.

Exam Strategy Summary

High-Yield Checklist for Game Theory (2/5 years, both most recent, 3–6 marks)

1. **Memorize the underline method:** for each column, underline the row-player's best (highest) payoff; for each row, underline the column-player's best (highest) payoff. Any cell with BOTH numbers underlined is a Nash Equilibrium — there can be 0, 1, or 2+.
2. **A 2×2 matrix can have TWO NEs** (coordination games like 2024 Eve/Mallory) — don't stop after finding one if it's a "prefer to match" scenario.
3. **Strictly dominant strategy** = one action is always best regardless of the opponent's choice. Check each player's payoffs for BOTH of their own actions against EACH of the opponent's actions — if one row/column choice wins in every comparison, it's dominant.
4. **If both players have the same dominant strategy** and it coincides with the unique NE (2023 Alice/Bob, both ST#1), explicitly say so — this is the "cleanest" answer and often worth bonus clarity marks.
5. **Conceptual variant** (2023 Q4(b)): "design a strategy" $\rightarrow$ describe the server/node model (actions, payoffs, surveillance trade-off) and conclude with "find the Nash equilibrium of the payoff matrix to balance security vs. service quality."
6. **Always show the labeled payoff matrix with underlines** — this is the visual proof that earns full method marks even if the final NE identification has a typo.